

Tehreza Uenarez COM 221 August 27, 2023 Exercise 4

- Step 1
- $V_{k+1}(A) = \frac{1}{4} \cdot [(-1 + V(A)) + (-1 + V(B)) + (-1 + V(C)) + (-1 + V(D))]$
 $V_{k+1}(A) = \frac{1}{4} \cdot [(-1 + 0) + (-1 + 0) + (-1 + 0) + (-1 + 0)]$
 $V_{k+1}(A) = -1$
 - $V_{k+1}(B) = \frac{1}{4} \cdot [(-1 + V(A)) + (-1 + V(B)) + (-1 + V(E)) + (-1 + V(D))]$
 $V_{k+1}(B) = \frac{1}{4} \cdot [(-1 + 0) + (-1 + 0) + (-1 + 0) + (-1 + 0)]$
 $V_{k+1}(B) = -1$
 - $V_{k+1}(D) = \frac{1}{4} \cdot [(-1 + V(D)) + (-1 + V(E)) + (-1 + V(A)) + (-1 + V(C))]$
 $V_{k+1}(D) = \frac{1}{4} \cdot [(-1 + 0) + (-1 + 0) + (-1 + 0) + (-1 + 0)]$
 $V_{k+1}(D) = -1$
 - $V_{k+1}(E) = \frac{1}{4} \cdot [(-1 + V(B)) + (-1 + V(D)) + (-1 + V(A)) + (-1 + V(C))]$
 $V_{k+1}(E) = \frac{1}{4} \cdot [(-1 + 0) + (-1 + 0) + (-1 + 0) + (-1 + 0)]$
 $V_{k+1}(E) = -1$
 - $V_{k+1}(F) = \frac{1}{4} \cdot [(-1 + V(E)) + (-1 + V(D)) + (-1 + V(A)) + (-1 + V(C))]$
 $V_{k+1}(F) = \frac{1}{4} \cdot [(-1 + 0) + (-1 + 0) + (-1 + 0) + (-1 + 0)]$
 $V_{k+1}(F) = -1$
 - $V_{k+1}(H) = \frac{1}{4} \cdot [(-1 + V(G)) + (-1 + V(I)) + (-1 + V(J)) + (-1 + V(E))]$
 $V_{k+1}(H) = \frac{1}{4} \cdot [(-1 + 0) + (-1 + 0) + (-1 + 0) + (-1 + 0)]$
 $V_{k+1}(H) = -1$
7. Put the new value
- | | | |
|----|----|----|
| -1 | -1 | 0 |
| -1 | -1 | -1 |
| 0 | -1 | 0 |
- $q_{k+1}(A, \text{LEFT}) = -1 + V(A)$
 $q_{k+1}(A, \text{LEFT}) = -1 + (-1)$
 $q_{k+1}(A, \text{LEFT}) = -2$
 - $q_{k+1}(A, \text{RIGHT}) = -1 + V(B)$
 $q_{k+1}(A, \text{RIGHT}) = -1 + (-1)$
 $q_{k+1}(A, \text{RIGHT}) = -2$
 - $q_{k+1}(A, \text{UP}) = -1 + V(D)$
 $q_{k+1}(A, \text{UP}) = -1 + (-1)$
 $q_{k+1}(A, \text{UP}) = -2$

B. $qk+1(D, LEFT) = -1 + (-1)$
 $qk+1(D, LEFT) = -2$
 $qk+1(D, RIGHT) = -1 + (0) + (0)$
 $qk+1(D, RIGHT) = -1$
 $15. qk+1(C, UP) = (-1) + (-1) + (-1)$
 $qk+1(C, UP) = -3$
 $16. qk+1(C, DOWN) = -1 + (-1)$
 $qk+1(C, DOWN) = -2$
 $17. \pi k+1(B) = RIGHT$
 $18. qk+1(D, LEFT) = -1 + (-1)$
 $qk+1(D, LEFT) = -2$
 $19. qk+1(D, RIGHT) = -1 + (-1)$
 $qk+1(D, RIGHT) = -2$
 $20. qk+1(C, UP) = (-1) + (-1) + (-1)$
 $qk+1(C, UP) = -3$
 $21. qk+1(D, DOWN) = -1 + (0)$
 $qk+1(D, DOWN) = -1$
 $22. \pi k+1(D) = DOWN$
 $23. qk+1(E, LEFT) = -1 + (-1)$
 $qk+1(E, LEFT) = -2$
 $24. qk+1(E, RIGHT) = -1 + (0)$
 $qk+1(E, RIGHT) = -1$
 $25. qk+1(E, UP) = -1 + (-1)$
 $qk+1(E, UP) = -2$
 $26. qk+1(E, UP) = -1 + (-1) + (-1)$
 $qk+1(E, UP) = -3$
 $27. \pi k+1(E) = LEFT, UP, RIGHT, DOWN(A)$
 $28. qk+1(F, LEFT) = -1 + (-1)$
 $qk+1(F, LEFT) = -2$
 $29. qk+1(F, RIGHT) = -1 + (-1)$
 $qk+1(F, RIGHT) = -2$

30. $qk+1(F, UP) = -1 + (0)$
 $qk+1(F, UP) = -1$
 31. $qk+1(F, DOWN) = -1 + (0) + (-1)$
 $qk+1(F, DOWN) = -2$
 32. $\pi k+1(CF) = UP, DOWN$
 $33. qk+1(H, LEFT) = -1 + (0)$
 $qk+1(H, LEFT) = -1$
 $34. qk+1(H, RIGHT) = -1 + (0) + (-1)$
 $qk+1(H, RIGHT) = -2$
 $35. qk+1(H, UP) = -1 + (-1)$
 $qk+1(H, UP) = -2$
 $36. qk+1(H, DOWN) = -1 + (-1)$
 $qk+1(H, DOWN) = -2$
 37. $\pi k+1(V) = LEFT, RIGHT$
 38. $qk+1(V) = LEFT, RIGHT$
 39. $V*(A) = \frac{1}{4} \{ (-2) + (-2) + (-2) + (-2) \}$
 $V*(A) = -2$
 40. $V*(B) = \frac{1}{4} \{ (-2) + (-2) + (-2) + (-2) \}$
 $V*(B) = -2$
 41. $V*(D) = \frac{1}{4} \{ (-2) + (-2) + (-2) + (-2) \}$
 $V*(D) = -2$
 42. $V*(E) = \frac{1}{4} \{ (-2) + (-2) + (-2) + (-2) \}$
 $V*(E) = -2$
 43. $V*(F) = \frac{1}{4} \{ (-2) + (-2) + (-2) + (-2) \}$
 $V*(F) = -2$
 44. $V*(H) = \frac{1}{4} \{ (-2) + (-2) + (-2) + (-2) \}$
 $V*(H) = -2$

211

Phonetic Legend: COM 211

45. $q \# (A \# B) = -1 + (-2) = -3$

$r_1 \# -1 + (-1.75) = -2.75$

$0_1 \# -1 + (-2) = -3$

$D_1) = -1 + (-1.75) = -2.75$

$46. q \# (B \# C) = -1 + (-2) = -3$

$r_1 \# -1 + (0) = -1$

$0_1 \# -1 + (-1.75) = -2.75$

$D_1) = -1 + (-2) = -3$

~~$47. q \# (C \# A) = -1 + (-1.75) = -2.75$~~

~~$r_1 \# -1 + (0) = -1$~~

~~$0_1 \# -1 + (-2) = -3$~~

~~$D_1) = -1 + (-1.75) = -2.75$~~

$48. q \# (C \# B) = -1 + (-1.75) = -2.75$

$r_1 \# -1 + (-2) = -3$

$0_1 \# -1 + (-2) = -3$

$D_1) = -1 + (-2) = -3$

$49. q \# (C \# E) = -1 + (-1.75) = -2.75$

$r_1 \# -1 + (-1.5) = -2.5$

$0_1 \# -1 + (-1.75) = -2.75$

$D_1) = -1 + (-1.5) = -2.5$

$50. q \# (E \# A) = -1 + (-2) = -3$

$r_1 \# -1 + (-1.5) = -2.5$

$0_1 \# -1 + (0) = -1$

$D_1) = -1 + (0) = -1$

$51. q \# (H \# A) = -1 + (0) = -1$

$r_1 \# -1 + (-2) = -3$

$0_1 \# -1 + (-1.5) = -2.5$

$D_1) = -1 + (-1.5) = -2.5$

$52. \pi \# (A) = \text{RIGHT, DOWN}$

$55. \pi \# (F) = \text{UP, DOWN}$

$56. \pi \# (H) = \text{LEFT, RIGHT}$

$57. \pi \# (D) = \text{DOWN}$

$58. \pi \# (E) = \text{RIGHT, DOWN}$

MICS

211

Phonetic Legend: COM 211

57. -2

-1.75

-2

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

0

-1.5

MICS